A complete finite Cantor subgroup counterexample

Lean disproves universal finite cyclic-subgroup smoothness equality in the complete actual polynomial-pair group for $y^2 = x^5 + 1$ over F3. The proper subgroup generated by $(X, 1)$ has order 5, different from characteristic 3, and every member has split u. Its fraction differs from the ambient fraction under both identity conventions. [pcc:unverified]

Exact kernel-proved population	All pairs	Split u	Fraction
Ambient, including identity	10	8	4/5
Proper cyclic subgroup, including identity	5	5	1
Ambient, excluding identity	9	7	7/9
Proper cyclic subgroup, excluding identity	4	4	1

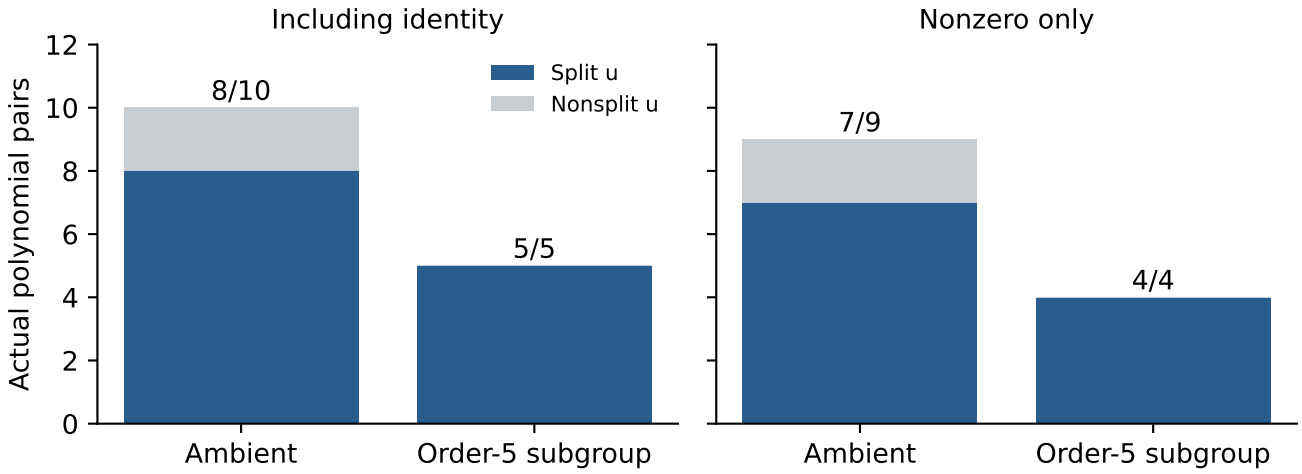


Figure 1: Exact finite counts from the main certificate and nonzero-domain lemmas. No sampling, uncertainty interval or runtime measurement.

The whole tracked project gate passes **808 declarations**, zero failures, zero trusted-native exceptions. Every one of the 100 ordered additions has an explicit polynomial-run certificate. Transitive axioms are restricted to propext, Classical.choice and Quot.sound. No custom mathematical axiom, hole or native decision exception is accepted. Lean 4.35.0 and Mathlib revision prefix `56a00585` are pinned. [pcc:unverified]

The full model ranges over all actual monic polynomials u of degree at most 2, with degree v below degree u and u dividing $v^2 - f$. Completeness is proved from all 91 bounded coefficient shapes, followed by a bijection to the ten valid pairs. The curve polynomial is monic, degree 5 and separable; splitting uses actual Polynomial.Splits. [pcc:unverified]

Arithmetic precedes group transport

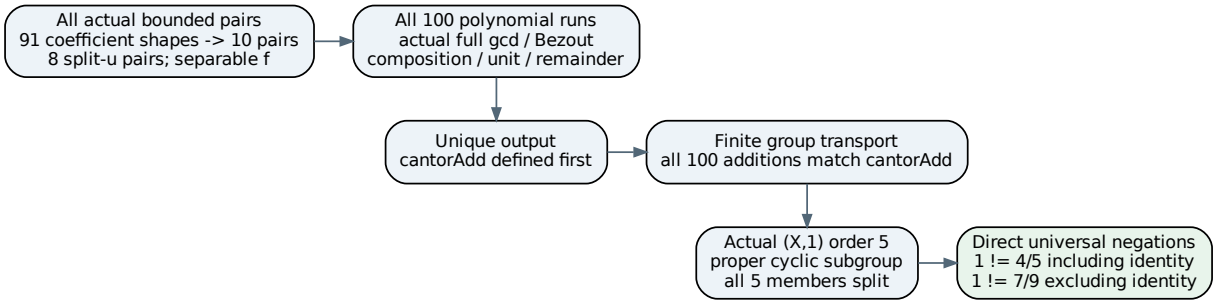


Figure 2: Proof dependencies. The unique polynomial output is defined before transferring the finite group.

The fixed Bezout choices certify the actual normalized full gcd. Each run checks exact composition and numerator division, bounded remainder, and either a low-degree output or a high-degree reduction with a nonzero normalization unit and exact final remainder. Generic cancellation and monic remainder uniqueness determine the output. `cantorAdd` is defined from those existence/uniqueness proofs before group transport. All 100 transported additions then match the actual run. [pcc:unverified]

The direct Lean negations cover both sampling domains:

- `universal_cyclic_fraction_equality_false`;
- `universal_nonzero_cyclic_fraction_equality_false`.

The combined binding is `full_stoll_counterexample_certificate`. A separate kernel negative control proves that omitting the normalization factor makes a concrete reduction equation false. This checks a weakened formula; the qualified Cantor algorithm is preserved. [pcc:unverified]

Source commit `a88c6fa0` precedes theorem packet `PCC-20261110-4d413c`. Exact declaration/source hashes, the full gate archive and failed attempts are linked from `../../full-stoll-sources.json` and `../../FULL_STOLL.md`. The local packet verdict and unsigned standing are recorded separately; scientific standing requires an independent signed verdict. [pcc:unverified]

The native candidate record remains separate and discloses contradictory frozen generic parameter labels. No full native protocol-conformance claim is made, and original raw kernel_proof_status remains unchanged. These Lean equations do not depend on the native run's acceptance. Primary Stoll 2014 Theorem 4.18 and Example 4.19 were read at the scopes recorded in stoll-sources.json; original Cantor 1987 remains unread after 403. [pcc:unverified]

The geometric Picard/Jacobian identification, a faithful scaling family, and qualified typical-subgroup or fixed-genus asymptotic claims remain separate. This is a complete finite arithmetic counterexample, not a refutation of Gaudry's qualified limit. The original fourteen-item audit is still partial. No new paid compute or novelty claim. [pcc:unverified]